

Analysis of Pathways for Optimizing Care Services for Older Adults with Cognitive Impairment — A Case Study of Practical Mini-Program Development

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Abstract

To address practical challenges including insufficient public awareness of elderly cognitive health, inconvenient early screening, inadequate social support for family caregivers, and low resource utilization of professional care institutions, this study develops a mini-program for elderly cognitive impairment care services based on the real construction context of dementia-friendly communities in Xuhui District, Shanghai. The mini-program integrates six core modules: online simple screening, brain health science popularization, online intervention courses, application for family care beds, introduction and application of professional institutions, and mutual-aid communities for caregivers. User feedback was collected during trial operation to implement multiple rounds of functional iteration and optimization. The results show that the mini-program can effectively lower the threshold for early cognitive impairment screening, relieve the physical and mental stress of family caregivers, build an online connection channel linking communities, elderly care institutions and medical resources, and provide digital support for the elderly with cognitive impairment and their caregivers.

Keywords: Cognitive Impairment; Elderly Care; Mini-Program; Community Screening; Social Support

According to data released by the National Bureau of Statistics, as of the end of 2023, the population aged 60 and above in China stood at 297 million, accounting for 21.1% of the total population; the population aged 65 and above numbered 217 million, representing 15.4% of the total. Against the backdrop of a rapidly deepening aging trajectory in China, the population living with cognitive impairment is expanding at an accelerating pace, and the issue of cognitive impairment has garnered escalating public attention, emerging as a salient factor compromising the health and well-being of the country's older adult population.

"Cognitive impairment" (abbreviated as CI) denotes a pathological process characterized by dysfunction in the higher cortical processing functions of the brain, which gives rise to deficits in learning and memory, accompanied by a spectrum of manifestations including aphasia, apraxia, agnosia, and impaired executive function. A chronic condition predominantly affecting the elderly, cognitive impairment follows a progressive degenerative course that culminates in dementia in its advanced stages.

China is home to the world's largest population of individuals with cognitive impairment and also registers one of the fastest growth rates globally. As documented in the *Report on the Current Status and Development of Care Services for Older Adults with Dementia*, over 180 million older adults in China suffer from chronic diseases, and the number of elderly patients with dementia aged 60 and above is estimated at approximately 15.07 million, with demand for cognitively impaired elder care services expanding rapidly. Projections indicate that by 2030, the population of older adults with dementia in China will reach 22.2 million, rising to 28.98 million by 2050—equivalent to roughly 1 in every 36 Chinese individuals. Accordingly, the exploration of a comprehensive care service system for older adults with cognitive impairment—coupled

with the construction of a robust social support infrastructure and the provision of urgently needed care services for affected seniors and their families—represents a research endeavor of substantial practical import and theoretical value.

This study takes Xuhui District, the central urban area of Shanghai, as the research object. It has conducted field investigations in four aging communities, five elderly care institutions and three community elderly service centers in Xuhui District, held interviews with 11 professional social workers, distributed 200 questionnaires and collected and organized 196 valid responses, and systematically reviewed the achievements and existing problems of the current government-led care services for the elderly with cognitive impairment in Shanghai.

I. Analysis of the Construction of an Elderly Cognitive Impairment-Friendly Urban District in Xuhui District

1. Status of the Elderly Population

As of the end of 2023, the registered elderly population aged 60 and above in Xuhui District totalled 348,200 persons, constituting 36.69% of the district's total population; those aged 65 and above reached 276,500 persons, or 29.14%, confirming that the district entered a state of profound aging long ago. Statistical data indicate a dementia prevalence rate of 15.5% among the population aged 60 and above, with the total number of older adults living with cognitive impairment in the district exceeding 50,000.

2. Practical Accomplishments

Xuhui District has advanced the development of cognitive impairment-friendly communities along six principal axes: First, public awareness and education: through community-based outreach and educational initiatives, the district disseminates knowledge concerning cognitive

impairment and its deleterious effects, fosters accurate public understanding of cognitive decline in later life, cultivates societal empathy and support for affected older adults and their families, and nurtures a harmonious and inclusive community milieu. Second, risk screening: leveraging the Shanghai Elderly Cognitive Impairment Risk Self-Testing System—a dedicated mobile application—the district administers cognitive impairment risk assessments for community-dwelling older adults, thereby establishing a baseline data repository of the community's cognitively impaired elderly population. Third, early intervention: the district delivers cognitive health promotion programs for community-dwelling older adults and provides non-pharmacological intervention training and early-stage care services for seniors with cognitive impairment. Fourth, family support: through caregiver skills training programs and mutual-aid networks for affected families, the district extends psychological support, capacity-building opportunities, and respite services to families of older adults with cognitive impairment. Fifth, resource brokerage. In partnership with community health service centers, the district mental health center, and other medical institutions, the district facilitates medical referrals for individuals requiring clinical diagnosis and treatment, and for those in need of ongoing care, it establishes linkages to institutional care and specialized medical resources. Sixth, platform development. Building upon existing infrastructure such as the "Life Box" community comprehensive elder care centers, the district has established Community Elderly Cognitive Impairment Support Centers, which function as coordinating platforms for rational resource allocation and the cultivation of sustainable community-based support mechanisms.

II. Current Challenges

Through focus group discussions and individual interviews conducted

during our fieldwork, we identified four principal deficiencies in the development of cognitive impairment-friendly communities in Xuhui District:

1. Inadequate Public Awareness of Cognitive Impairment Among Older Adults

The stigma surrounding cognitive impairment remains deeply entrenched. Compounded by an array of contributing factors, the screening uptake rate among older adults in Xuhui District stands at less than 10%, and among seniors identified as high-risk, the rates of clinical consultation and treatment uptake fall below 30%—markedly lower than comparable figures in developed nations. A considerable number of families still perceive early-stage cognitive decline as a normative aspect of the aging process, recognizing the gravity of the condition only when it has advanced to moderate or severe stages, at which point therapeutic interventions yield markedly diminished efficacy.

2. Limited Diversity of Early Screening Modalities for Cognitive Impairment

The scale-based assessment tools most commonly deployed for early cognitive screening impose certain professional qualification requirements, meaning that older adults in need frequently require assistance to complete the evaluation. Owing to the constrained pool of qualified specialists, screening campaigns conducted in partnership with professional service providers struggle to deliver simultaneous early screening services across all screening sites district-wide. While Xuhui District has piloted voiceprint-based screening technology this year, the constrained range of early screening channels and modalities continues to limit accessibility for older adults, and broader population-level penetration remains modest.

3. High Vacancy Rates in Cognitive Impairment Care Facilities

Cognitive impairment service facilities fall into two principal categories: First, residential care facilities with dedicated dementia care beds. Citywide, the number of dementia-designated care beds has reached 12,000, with Xuhui District accounting for over 1,000 beds; yet the actual occupancy rate dips below 50%. Pursuant to the government's annual mandatory bed-expansion targets, the total bed count is set to rise further to 15,000 by the conclusion of the *14th Five-Year Plan* period. The capital expenditure and operational costs associated with dementia-specific beds exceed those of standard elder care beds by 30% to 50%; the confluence of chronically low utilization rates and continued annual capacity additions engenders substantial resource misallocation and waste. Second, community day-care centers. Our field research reveals that community-based cognitive impairment day-care facilities across the city are typically operated by third-party providers, with an average capacity of approximately 15 slots, staffed by 3 to 4 caregivers, and supported by annual government operating subsidies ranging from 350,000 to 500,000 yuan. A confluence of factors—including insufficient public outreach, limited transportation accessibility, and prevailing stigma—dampens demand for day-care utilization, resulting in persistently elevated vacancy rates across many community facilities.

4. Inadequate Support for Families of Older Adults with Cognitive Impairment

Currently, upwards of 80% of older adults with cognitive impairment in China reside with family members. While home-based care affords a degree of security and continuity for patients, it simultaneously imposes multifaceted burdens on caregiver families. As cognitive capacities decline, the activities of daily living of affected seniors deteriorate apace, requiring family caregivers to acquire foundational caregiving knowledge and competencies. Concomitantly, family caregiving exacts a toll

extending well beyond time and physical exertion; primary caregivers must also navigate profound emotional and psychological distress. Throughout our fieldwork, we observed stress as the most pervasive challenge confronting caregivers. In the absence of formal social support infrastructure or shared caregiving responsibilities distributed among family members, primary caregivers frequently manifest a spectrum of psychological distress symptoms—including anxiety, irritability, depressive mood, feelings of frustration, and a pervasive sense of helplessness.

III. Practical Explorations of the Research Team

In direct response to the aforementioned gaps in cognitive impairment care and support, we have piloted a series of innovative practical interventions in the course of our community-based practice and volunteer service activities, in a concerted effort to identify avenues for systemic improvement. Following an exploratory phase, the research team has concentrated its efforts on digital technology enablement, seeking to expand resource linkages and deliver accessible, user-centric services to older adults and their families through the development and deployment of an Elderly Cognitive Impairment Service Mini-Program.

1. Problems the Mini-Program Seeks to Address

- (1) The challenge of limited public awareness. Devising innovative public education strategies to assist community residents in overcoming the stigma attendant to cognitive impairment and to heighten early risk recognition among older adults.
- (2) The challenge of screening inaccessibility. Enabling older adults to complete preliminary cognitive screening in the comfort of their own homes, thereby augmenting convenience, facilitating early detection and intervention, and mitigating the risk of delayed treatment.

(3) The challenge of insufficient support. Aggregating diverse service resources, establishing connections to professional institutions, delivering online family caregiver training programs, and enhancing caregiving competencies and psychological well-being through structured training and peer community support.

(4) The challenge of fragmented information: consolidating the full spectrum of medical, care, and daily-living service resources within Xuhui District into a unified platform, enabling older adults in need and their families to readily access requisite service information and resources, and thereby enhancing overall service accessibility.

2. Core Modules of the Mini-Program

(1) Online cognitive screening. Built upon the district's unified risk screening metrics, a validated online cognitive impairment screening module will be developed. Users may assess their cerebral health status by responding to a structured questionnaire and receive an instant preliminary screening report. For individuals exhibiting potential risk, the system furnishes tailored recommendations for further clinical evaluation, thereby ensuring timely detection and intervention.

(2) Brain health education and awareness. Through richly illustrated articles, instructional video series, and multimedia formats, the mini-program disseminates evidence-based knowledge of cognitive impairment to community residents—encompassing symptom recognition, preventive measures, and therapeutic approaches. Content is updated on a regular basis, and expert-led lecture series are featured to elevate public literacy concerning cognitive health.

(3) Online intervention course instruction. A dedicated online instructional module will be established, spanning memory training, language communication techniques, and related domains. Under the clinical supervision of the municipal and district mental health centers, a

standardized non-pharmacological intervention curriculum will be finalized, yielding a rigorous, practical, user-friendly, and evidence-based course framework and intervention protocol. Residents may select courses aligned with their individual needs to progressively enhance cognitive function.

(4) Referral and application for home-based dementia care beds. For seniors requiring in-home care services, users may directly browse nearby professional home care bed services through the mini-program and submit applications for home-based cognitive impairment care beds.

(5) Referral and application for day-care and institutional care services
Recognizing that certain families may be unable to provide around-the-clock supervision for individuals with cognitive impairment, the mini-program will feature dedicated application portals for day-care services and specialized dementia care facilities. Institutional services are presented through video tours, photographic galleries, and descriptive text, enabling residents to apply for specific care services as required.

(6) Establishment of family caregiver communities

An online peer community comprising family members of individuals with cognitive impairment will be inaugurated to provide both emotional sustenance and practical guidance. Family caregivers may exchange experiences and insights, while also participating in regularly scheduled caregiver training workshops to refine their caregiving practice.

3. Architectural Design of the Mini-Program

Drawing upon the identified challenges and the envisaged service content, the research team has developed a foundational architectural blueprint for the mini-program and is engaged in the ongoing enrichment of its constituent modules and functionalities. The overall architecture is depicted in the figure below:

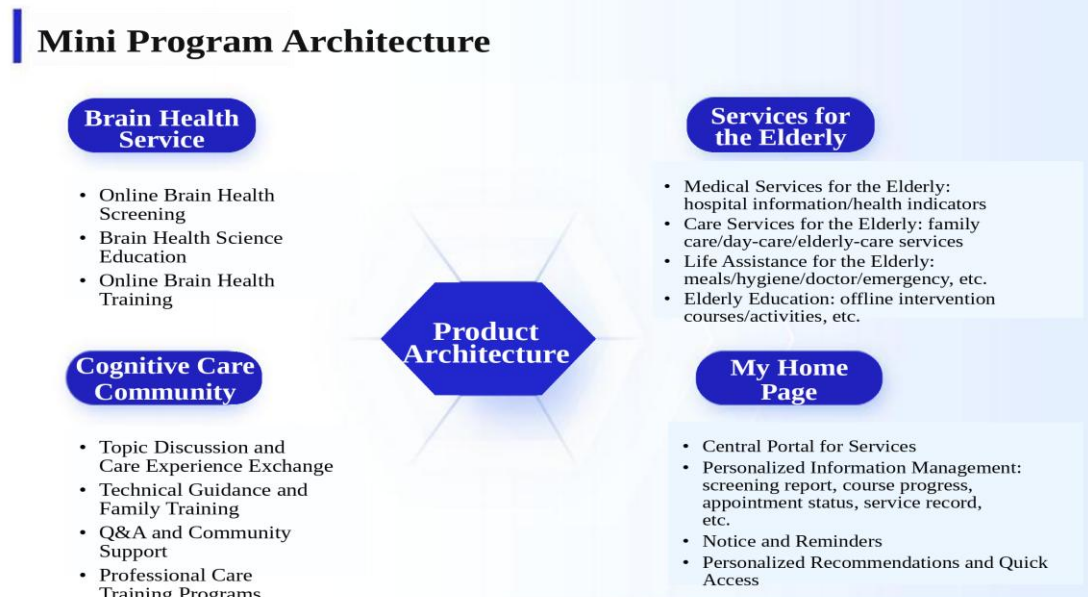


Figure 1 Architecture Diagram of the Elderly Cognitive Impairment Service Mini-Program

4. Core Functions of the Mini-Program

Cognitive screening: Delivering precise assessments of cognitive health with individualized report generation.

Public education: Multi-modal dissemination of brain health literacy to augment user awareness.

Intervention training: Personalized cognitive training regimens to enhance users' cognitive capacities.

Elderly services: Integrated service ecosystem encompassing medical care, custodial care, and daily-living assistance.

Community interaction: Delivering technical guidance and emotional support for family caregivers.

5. Interface Design of the Mini-Program

With the professional guidance of industry experts, the research team has designed and prototyped Version 1.0 of the mini-program. Through a

clean, intuitive interface design and intelligent feature set, the platform aims to enable senior users to effortlessly access the services they require.

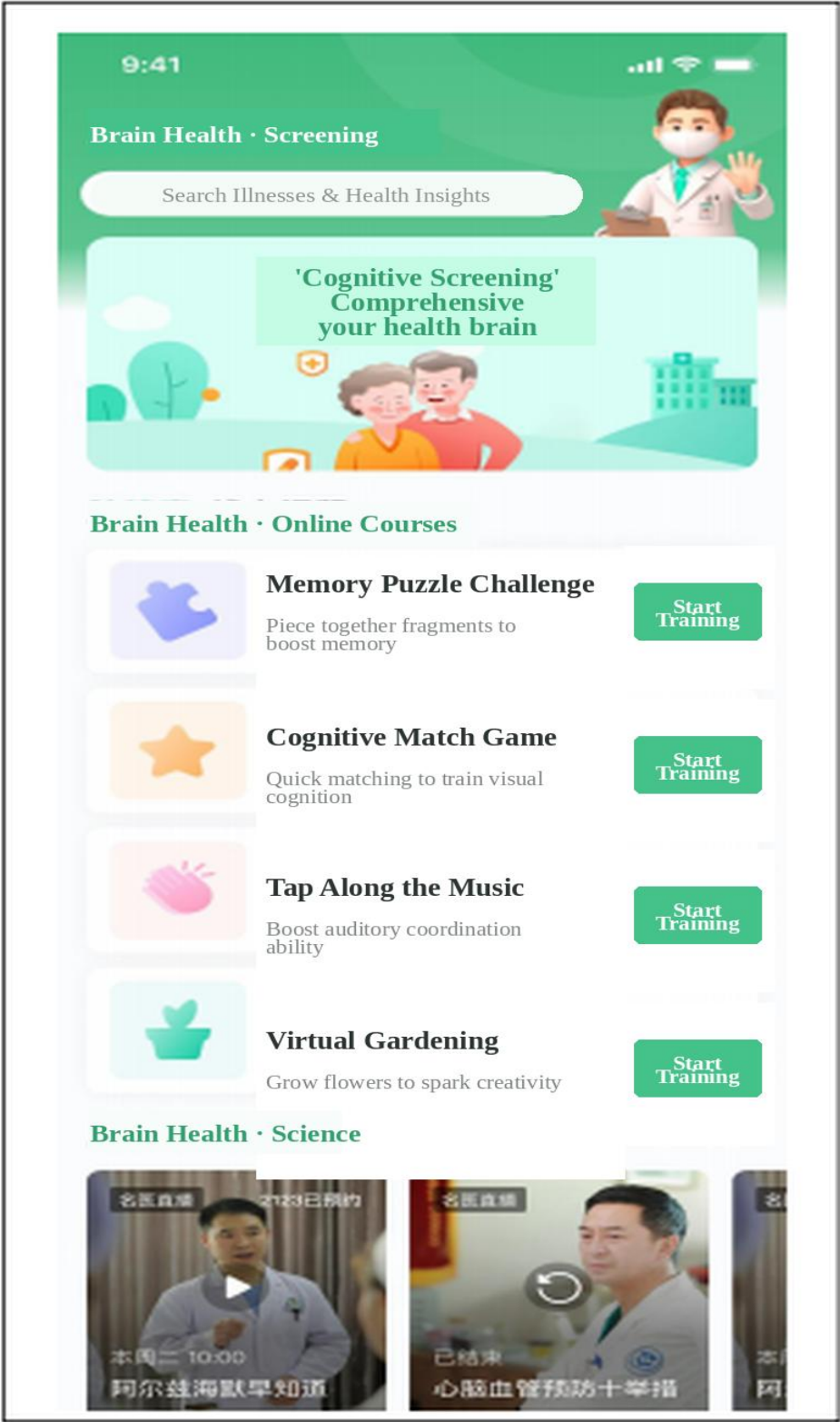


Figure 2 Interface of the Elderly Cognitive Impairment Service Mini-Program — Brain Health Services



Figure 3 Interface of the Elderly Cognitive Impairment Service Mini-Program — Services for Older Adults

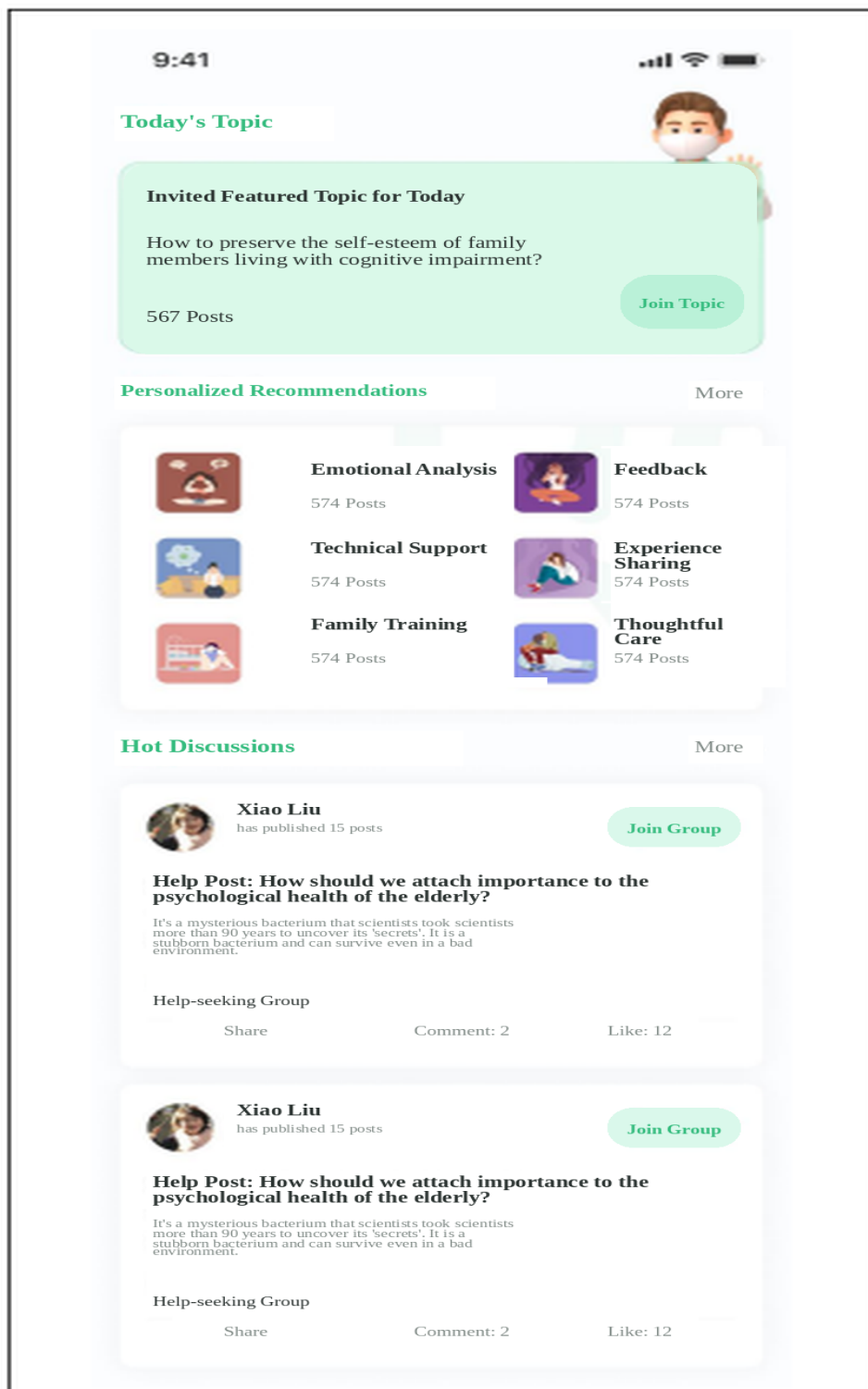


Figure 4 Interface of the Elderly Cognitive Impairment Service Mini-Program — Cognitive Care Community

6. Pilot Evaluation of the Mini-Program

With the collaborative support of social workers from three communities in Xuhui District—Lingyun Sub-district, Changqiao Sub-district, and Huajing Town—the research team recruited 68 older adult participants to evaluate Version 1.0 of the mini-program. Following the pilot period, participants furnished comments and suggestions, which are consolidated as follows:

Category of Feedback	Specific Comments / Respondent Count (Percentage)
Online screening	1. Prefer a shorter completion time; the current duration of 8 minutes should ideally be reduced to under 5 minutes; 2. Following the provision of diagnostic recommendations, users wish to receive information on nearby medical examination facilities and, ideally, the option of online appointment booking. 11 respondents / 16% 8 respondents / 12%
Functional modules	1. While the core functions are largely in place, the substantive content within each module requires further enrichment; 2. Within the "Geriatric Medical Care" module, the green-channel and specialist-finder features are presently limited to community health stations, where queuing is generally not a concern; users request the addition of online consultation and appointment booking with specialists from higher-tier hospitals; 3. The instructional videos on family caregiving could be more comprehensive and detailed. 21 respondents / 31% 3 respondents / 5% 17 respondents / 25%

Interface design	1. The font size is too small, necessitating a magnifying glass to read comfortably; 2. The interface feels cluttered; desired functions are difficult to locate; a more streamlined layout is preferred. 23 respondents / 34% 13 respondents / 19%
Data privacy	Older adults register and log in using their mobile phone numbers; respondents express strong concern that their personal data not be disclosed. 31 respondents / 46%

Incorporating the feedback and suggestions proffered by the participating older adults, the research team has initiated revisions and enhancements. During the forthcoming winter vacation, the pilot scope will be broadened to solicit input from a wider user base. With the institutional backing of departments including the Xuhui District Civil Affairs Bureau and sub-district administrative offices, optimization and refinement will be pursued along the following six axes:

1. Systematizing and cataloguing the free service provisions and subsidy assistance policies for older adults—particularly those with cognitive impairment—across all tiers of government (municipal, district, and sub-district/township levels within Shanghai and Xuhui District, and integrating policy query and matching functionalities within the mini-program;

2. Compiling volunteer service resources, community service initiatives, and mutual-aid support programs available within Xuhui District for the reference of older adults in need.

3. Given the infeasibility of direct integration with higher-tier hospital resources, the "green channel" and "specialist finder" modules will be realigned, with service offerings calibrated to available institutional capacity.

4. Continuously updating the directory and profile of professional service institutions within Xuhui District; supplementing telephone consultation with online appointment booking capabilities.

5. Expanding recommendations for home-based age-appropriate retrofitting and smart safety products, with particular emphasis on government-subsidized services offered by the Shanghai Municipal Government and Xuhui District Government, and establishing dedicated online application portals.

6. Refining the user interface design, with font sizing and layout to better suit the needs and preferences of older adult users.

IV. Research Conclusions

This study addresses practical challenges including fragmented community care for older adults with cognitive impairment, low awareness of government service provision and inadequate resource linkage, and investigates the functional design, resource integration and application pathways of a dedicated care service mini-program. It is found through practical exploration that this information-based mini-program acts as an effective solution to the difficulties in community care for cognitively impaired older adults. As a lightweight, easily promotable and low-cost platform, it can systematically pool diverse resources covering community medical services, elderly care, volunteer forces and family care, build an integrated online-offline service system, and effectively overcome the fragmentation, inefficiency and passivity of conventional community care. Moving forward, the platform can be further upgraded with expanded functional modules and broader coverage, access to additional medical and care resources, and tailored intervention plans for older adults with varying degrees of cognitive impairment.

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